Abdussamad Mohammed

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Education

Ain Shams University — Cairo, Egypt
Bachelor of Science in Mechatronics Engineering; GPA: 3.03 — *Oct 2020 – Jul 2025*

Experience

Embedded Systems & Model Based Engineer — Oct 2024 – Jul 2025
CoTune Remote IoT Control Lab - Ain Shams University — *Cairo, Egypt*
- Developed MATLAB/Simulink dynamic models and real-time control systems for mechatronic platforms, including an inverted pendulum, and generated deployable embedded code using Simulink Coder
- Developed a Hardware-in-the-Loop (HIL) architecture integrating Simulink with ESP32-based hardware for real-time execution, monitoring, and control of physical systems
- Implemented embedded interfaces for encoders, IMU sensors, motor actuators, and MQTT communication, enabling Simulink models to interact directly with physical hardware
- Developed IoT connectivity using ESP32, Wi-Fi, MQTT, and AWS IoT for remote experiment control, real-time telemetry, parameter configuration, and data/video feedback

Camera-Radar System Engineer — Oct 2024 – Jul 2025
Dronodat GmbH — *Remote (Germany)*
- Designed a camera mounting system for moving vehicles, prioritizing mechanical stability and field adjustability for inspection applications
- Researched and selected suitable radar modules based on performance and commercial availability
- Developed a GUI for camera parameter configuration using the Sony Remote Camera SDK to support rapid prototyping
- Investigated radar-camera sensor fusion approaches by reviewing academic literature and implementation strategies
- Explored methods for calibration and synchronization between radar and camera systems to enable accurate data fusion

UAV Camera Engineer — Nov 2023 – Jun 2024
Autonomous Research Lab - Ain Shams University — *Cairo, Egypt*
- Designed an aerial car detection algorithm using machine learning for object detection
- Built an interfacing system with the ZED camera for autonomous systems

Hardware & PCB Designer — Oct 2022 – Present
Freelance — *Cairo, Egypt*
- Designed embedded hardware and manufacturing-ready PCBs incorporating microcontrollers, motor drivers, sensors, power regulation, USB interfaces, and communication peripherals
- Developed complete hardware architectures from component selection and schematic design through PCB layout and manufacturing preparation

Hardware and Software Design Trainee — Sep 2022 – Feb 2023
Orange Digital Center - AMIT — *Cairo, Egypt*
- Trained on PCB design and embedded systems development using C/C++
- Built peripheral drivers for AVR microcontrollers

Business Development Specialist — Nov 2023 – Jun 2024
MECA - Ain Shams University — *Cairo, Egypt*
- Organized events and programs to prepare university students for the job market
- Collaborated with sponsors to execute successful student events

Presentation Committee Member — Nov 2023 – Jun 2024
Enactus - Ain Shams University — *Cairo, Egypt*
- Represented Enactus ASU in national and international events
- Joined the scriptwriting team for competitions and events

Business Logic Case Team Leader — Oct 2021 – Jul 2023
ASU Racing Team — *Cairo, Egypt*
- Led the business team in 2023 and authored the 2022 business plan
- Mentored team members on business concepts and competition strategy
- Team achieved 3rd Place Worldwide

Projects

Autonomous Production Line | *SOLIDWORKS, Python, C/C++, Raspberry Pi, MATLAB, Simulink*
- Built an automated production line using a SCARA robot based on VDI-Model methodology

Aerial Object Detection | *Python, OpenCV, Raspberry Pi, ZED Camera*
- Developed a machine learning model for object detection with ZED camera on UAV

Autonomous Car Simulation Model | *Python, PyTorch*
- Built a CNN-based controller for autonomous navigation in a simulated environment

Kalman Filter Implementation | *Python, ROS, Gazebo*
- Implemented a 2D Kalman filter to fuse noisy measurements and validated results

Racing Aeolus Model Car | *SOLIDWORKS, Arduino*
- Led the braking and control subsystems for a 1:2 scale racing car model
- Achieved 2nd place in the Racing Aeolus competition

Remote Controlled Batmobile | *SOLIDWORKS, Arduino, PCB Design*
- Designed a Batman-themed RC car with line-following, projectile shooting, and wireless gate communication

Technical Skills

Embedded & IoT: Embedded Systems, ESP32, Arduino, AVR, MQTT, AWS IoT, C/C++
Model-Based Engineering: MATLAB, Simulink, Simulink Coder, Hardware-in-the-Loop (HIL), PID Control
Hardware: PCB Design, Altium Designer, Sensors, Motor Drivers, Embedded Hardware Design
Programming: C/C++, Python, MATLAB
AI & ML: Machine Learning, Deep Learning, Neural Networks, PyTorch
Robotics: Forward/Inverse Kinematics, ROS, Gazebo, Raspberry Pi
Mechanical Engineering: Mechanical Design, SOLIDWORKS